Venkata Lakshmi Parimala Pasupuleti

AI Customer Engineer / Forward Deployed Engineering · Enterprise AI Deployments · Google Cloud · Vertex AI · Python · Agentic AI · Production Troubleshooting

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Virginia, USA · open to any US location

PROFESSIONAL SUMMARY

AI Customer Engineer with 8 years resolving complex technical issues in production enterprise systems — de-facto owner of technical incident resolution at Entrata (99.5% uptime, zero compliance violations, debugging tools reducing resolution time 40%). LeaseI: production 8-agent LangGraph enterprise AI deployment (enterprise integrations, compliance guardrails, HITL escalation, 93.6% precision across 1,240 test cases). AgentProbe: enterprise AI evaluation at scale (CI regression harness, LLM-as-judge, multi-provider model routing). GCP/Vertex AI; Gemini-equivalent LLM orchestration. Executive communication: 600+ student workshops, translating deep technical issues into business impact for enterprise stakeholders. PHI/HIPAA/SOC 2/NIST/OWASP LLM Top 10. M.S. Data Analytics Engineering, GMU (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- AI/ML model development, debugging production reliability issues, API integration; communicating technical issues to non-technical stakeholders; responsible AI and data governance practices.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- End-to-end enterprise AI solution deployment: LangGraph agentic platform (parse → plan → retrieve → validate → route), RAG over regulatory corpus, safety guardrails, enterprise API integration; translated implementation into stakeholder-facing outcomes.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Technical workshops to 600+ students (Python, SQL, OOP) translating complex concepts for diverse audiences; faculty research presentations to non-specialist leadership; one-on-one debugging support; ML prediction pipelines (Python, R) $R^2 = 0.977$.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **8 years resolving complex technical issues for customers and cross-functional partners** (Python, Java, PHP, PostgreSQL, RabbitMQ, AWS) — de-facto owner of technical incident resolution for 100K+ MAU enterprise platform (99.5% uptime); diagnosed and resolved production incidents end-to-end across unfamiliar systems; built internal debugging tools reducing resolution time 40%; maintained bug count below 10 for 6+ consecutive months; zero compliance violations across 7 years. Translated deep technical root causes into business-impact language for product stakeholders and account teams.
- **Enterprise integrations and cloud infrastructure** — led Affordable Housing module integration for AWS S3 enterprise-scale document migration (100K+ records, PHI-compliant, Libsodium-encrypted, zero security incidents); third-party REST API integrations and identity provider integration; multi-tenant architecture; OData APIs; 200+ SQL query optimizations (30% performance gain); Docker + Kubernetes microservices; CI/CD (GitHub Actions); AWS (S3, EC2, RDS, Lambda, Step Functions, API Gateway, IAM).
- **Executive communication and mentoring** — translated complex regulatory and technical constraints into stakeholder-facing solutions; field feedback translation into feature requests and product changes; mentored 5+ engineers; 300+ peer code reviews; collaborated with global teams across 120+ Agile sprints; Python/Bash automation eliminating 80% of manual workload.

PROJECTS

Leasel: Enterprise AI Deployment - 8-Agent LangGraph Compliance Platform
LANGGRAPH HITL

Python · LangGraph · FastAPI · Vertex AI-compatible · ChromaDB · MCP · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production enterprise AI: 8-agent LangGraph (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Escalation) with enterprise integrations, compliance boundaries (regulated decisions route to human review), citation-grounded outputs. Diagnosed novel implementation bugs at code level; translated root-cause findings to business-impact recommendations. **93.6% precision, 87.9% F1 across 1,240 compliance test cases** (5 US states + federal).

AgentProbe: Open-Source AI Evaluation Toolkit for Enterprise Deployments
LLM-AS-JUDGE MULTI-CLOUD

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Enterprise AI evaluation framework: 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate); LLM-as-judge; CI regression harness with golden datasets; multi-provider model routing (OpenAI/Anthropic/Groq/Ollama, Gemini-compatible); trace-level observability; offline + online evaluation. Used to validate LeaseI across 1,240+ test cases.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Customer Engineering & FDE: Troubleshooting Technical Issues (8yr), Enterprise AI Deployments, Enterprise-Grade AI Solutions, Pre-sales Support, Post-sales Technical Design, Technical Evaluation, Enterprise Integrations, API Design, REST APIs, Third-Party Integrations, Identity Provider Integration, Enterprise Content Management (ECM), Agent Connectors, Account Team Collaboration, Field Feedback Translation, Feature Request Prioritization, Executive Communication, Business Impact Translation, Solution Deployment, Customer Adoption, Technical Onboarding, Compliance Boundaries

GCP, Vertex AI & Agentic AI: Google Cloud Platform (GCP), Vertex AI, Gemini Models, Google Cloud Storage, LangGraph, LangChain, LangSmith, MCP, Multi-Agent Orchestration, Agentic AI, AI Agent Workforce, RAG Pipelines, Hybrid Search, Semantic Search, ChromaDB, Qdrant, OpenAI API, Anthropic API, Groq, Ollama, Multi-Provider LLM Integration, Prompt Engineering, HITL, Guardrails, OWASP LLM Top 10, Hallucination Detection, LLM-as-Judge, Trace-Level Observability — AWS (S3, EC2, RDS, Lambda, Bedrock, SageMaker, API Gateway, IAM), Azure AI Foundry, Azure OpenAI Service, Azure ML, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), FastAPI, PostgreSQL, PHI Compliance, HIPAA, SOC 2, NIST

Languages & ML: Python (8yr), Java (Spring Boot), TypeScript, JavaScript, SQL, Bash — PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, BERT, Model Deployment, MLOps, Multi-Tenant Architecture, Microservices, Distributed Systems, OpenTelemetry, Prometheus, Grafana